

1)

- a) List 4 legal identifier names
 - 1. numBeads
 - 2. playerScore
 - 3. totalHours
 - 4. myNumber
- b) List 4 illegal identifier names
 - 1. 2players: cannot start with a number
 - 2. My-number: cannot have hyphen
 - 3. class: cannot use Java keyword
 - 4. My number: cannot use spaces

2)

- a) In two statements, declare a variable named numBeads and assign it with the value 5
 - 1. `int numBeads;`
 - 2. `numBeads = 5;`
- b) In one statement, declare a variable named numBeads and assign it with the value 5
 - 1. `int numBeads = 5;`

3)

- a) What is the final value of yourNumber after the last statement executes?
`Int myNumber = 5;`
`Int yourNumber = 4;`
`myNumber = yourNumber * 2;`
`yournumber = myNumber + 5;`
`yourNumber = 13`
- b) What is the final value of yourNumber after the last statement executes?
`Int myNumber;`
`Int yourNumber = 4;`
`myNumber = yourNumber + 7;`
`yourNumber = myNumber;`
`yourNumber = 11`

4) Determine the appropriate data type for each of the following values:

- a) The number of basketballs in a department store
`int`
- b) The price of a basketball
`double`
- c) The number of players on a basketball team
`int`
- d) The average age of the players on a basketball team
`double`
- e) Whether a basketball player has received a jersey or not
`boolean`
- f) The first initial of a basketball player's first name
`char`

5)

- a) What is the difference between a primitive data type and an abstract data type?
A primitive data type is built directly into Java, such as int, double, boolean, char

An abstract data type is defined by the data it stores, such as list or queue.

- b) What is the difference between a class and an object?
A class is a blueprint that defines the properties of something

An object is an actual instant

E.g. if exercise is the class, then lesson is the object
exercise lesson = new exercise();

11)

Using the following declarations, rewrite the statements to include the appropriate type casting, rounding where necessary. If type casting is not necessary, explain why:

```
Int j = 5;  
Double k = 1.6;  
Int y;  
Double z;
```

- a) $y = j * k$;
 $y = (int)(j * k)$;
 $y = 5 \times 1.6$
 $y = 8$
- b) $z = j * k$;
No casting is necessary because z is a double
 $z = 8.0$
- c) $z = k * k$;
No casting is necessary because z is a double
 $z = 2.56$
- d) $j = k$;
 $j = (int)k$;
 $j = (int)1.6$;
 $j = 1$
- e) $k = j$;
No casting because k is a double
 $k = 5.0$
- f) $y = j + 3$;
No casting because they are all integers
 $y = 8$